

Matgeo-q.5.3.30

AI25BTECH11036-SNEHAMRUDULA

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Question. Solve the system of linear equations (using the matrix method)

$$7u + 2v = 11, \quad 4u - v = 2.$$

Solution.

$$\begin{pmatrix} 7 & 2 \\ 4 & -1 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 11 \\ 2 \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 7 & 2 \\ 4 & -1 \end{pmatrix}, \quad \mathbf{p} = \begin{pmatrix} u \\ v \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 11 \\ 2 \end{pmatrix}. \quad (1)$$

Since

$$\det(\mathbf{A}) = -15 \neq 0, \quad (2)$$

the unique solution is $\mathbf{p} = \mathbf{A}^{-1}\mathbf{b}$ with

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \begin{pmatrix} -1 & -2 \\ -4 & 7 \end{pmatrix} = \frac{-1}{15} \begin{pmatrix} -1 & -2 \\ -4 & 7 \end{pmatrix}. \quad (3)$$

Hence

$$\mathbf{p} = \frac{-1}{15} \begin{pmatrix} -1 & -2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 11 \\ 2 \end{pmatrix} = \frac{-1}{15} \begin{pmatrix} -15 \\ -30 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}. \quad (4)$$

Therefore,

$$u = 1, \quad v = 2. \quad (5)$$

and the number of unknowns is also 3, the system has a **unique solution**.

Graphical Representation

